Exam 1: RapidoMX Delivery Analytics

Regression Analysis · ITESO · Prof. María Elisa Vaca Gómez

The case: a memo from the COO

You are the new analytics team at **RapidoMX**, a food delivery app in Mexico. Three leaders want answers from the same order data, and you must give all three a clear answer backed by evidence.

> **To:** Analytics team
>
> **From:** Chief Operating Officer, RapidoMX
>
> **Subject:** Three questions before our quarterly planning
>
> Team, before we plan next quarter I need three answers. Each one comes from a different area, but they all use the same order data.
>
> **1. Operations asks: what makes a delivery slow?** We promise customers a delivery time in the app. We want to know which factors really drive delivery time, how many minutes each one adds, and whether a model is good enough to give customers a reliable estimate.
>
> **2. The Regional Director asks: are some zones worse than others?** Her report says Centro is our fastest zone and Periferia is our slowest. She proposes moving riders from Centro to Periferia next quarter. Before we move people, I want to know if the data supports this decision.
>
> **3. Customer Experience asks: who ends up with a bad experience?** A rating of 1 or 2 stars is a bad experience, and those customers often stop ordering. We want to understand what causes bad experiences. We also want to send a discount coupon to orders at risk **at the moment the order is placed**, before anything goes wrong.
>
> Please send me a short memo with your recommendations. I am not a statistician, so explain your answers in business terms.

The three questions are connected. What you learn in one part should inform the next ones, so read the whole exam before you start.

Exam rules and logistics

This is an individual take home exam. You have one week and should plan for about 7 to 8 hours of work.

Item	Detail
Deliverable	One Jupyter notebook (.ipynb) with all parts, the decision log and the executive memo.
File name	Exam_RapidoMX_Analysis_Regression_<StudentName>.ipynb

Train/test split. Use 70% train and 30% test with random_state=42.

What is expected in every analysis

These rules apply to the whole exam and are graded in every part.

- **Hypothesis tests.** For every test you run, write H_0 and H_1 in words, applied to this problem. Report the statistic and p-value, state your decision with $\alpha = 0.05$, and interpret what it means for the model and for the business.
- **Assumptions.** Every model you use has assumptions. Check them. When one does not hold, explain what it affects and how much you can still trust your conclusions.
- **Interpretation.** Interpret every result in business terms, with units (minutes, pesos, percentages, etc.). A number without interpretation earns no points.
- **Categorical variables.** Always name the reference category. Use zona = Norte, tipo_vehiculo = Moto, franja_horaria = Mañana and clima = Despejado.
- **Decisions.** Justify every **important decision** (cleaning, variables, methods, etc.) in your decision log.

The data

The file rapidomx_pedidos.csv has one row per order: 14 variables taken directly from the operational system of the app. Each row is one delivered order in Mexico City.

Variable	Type	Description
id_pedido	ID	Unique order identifier
zona	Categorical	City zone of the delivery address: Centro, Norte, Sur, Periferia
franja_horaria	Categorical	Time of day the order was placed: Mañana, Comida, Tarde, Noche
clima	Categorical	Weather when the order was placed: Despejado (clear), Lluvia (rain)
distancia_km	Continuous	Distance from the restaurant to the customer, in km
pedidos_activos_zona	Count	Number of active orders in the zone when the order was placed (how busy the zone is)
tipo_vehiculo	Categorical	Vehicle of the assigned rider: Moto, Bicicleta, Auto
experiencia_repartidor_meses	Continuous	Months the assigned rider has worked for RapidoMX
tiempo_preparacion_min	Continuous	Restaurant's estimated preparation time, in minutes, shown when the order is accepted
monto_pedido_mxn	Continuous	Order amount, in Mexican pesos
cliente_nuevo	Binary	1 if this is the customer's first order, 0 otherwise
tiempo_entrega_min	Continuous	Minutes from order placed to order delivered, recorded when the rider marks the order as delivered
pedido_con_error	Binary	1 if the customer reported a missing or wrong item after delivery, 0 otherwise
calificacion_cliente	Ordinal	Rating the customer gave after delivery, from 1 (worst) to 5 (best)

Warning: the data has not been cleaned. It comes straight from the operational system. Finding the problems is part of the exam.

Part 1. Data preparation and exploration (15 points)

Before you answer anyone, you need data you can trust. Prepare a clean dataset and document every problem you found, what you did about it and why.

Then explore the data with the three questions in mind. Your exploration should guide the rest of your analysis: close this part with **at least one finding for each leader** that you will check later in the exam.

Part 2. Operations: what makes a delivery slow? (35 points)

Operations wants to know what drives delivery time, how many minutes each factor adds, and whether a model is reliable enough to show customers an estimated delivery time.

Build a multiple linear regression model for tiempo_entrega_min. You choose the predictors, but you must justify your choice. Your model must include zona, because you will use it in Part 3. Your analysis should show:

- whether the model as a whole and each predictor are significant.
- whether the model's assumptions hold, and what that means for your conclusions.
- what each relevant factor means for delivery time, in minutes.
- how well the model predicts on data it has not seen, using all the performance metrics we saw in class, each one interpreted.

Close with a clear answer for Operations: which factors matter most, which ones RapidoMX can act on, and how much margin an estimated delivery time based on this model could have.

Part 3. Regional Director: are some zones really worse? (20 points)

The Regional Director's report says Centro is the fastest zone and Periferia the slowest, and she wants to move riders from Centro to Periferia next quarter.

Use ANOVA to test whether delivery times differ between zones. Then check which pairs of zones differ, using the method we saw in class. Then compare these results with what your Part 2 model says about the zones. If the two analyses tell different stories, explain why, with evidence from your data.

Close with a clear recommendation: should RapidoMX move riders from Centro to Periferia and why?

Part 4. Customer Experience: who ends up with a bad experience? (25 points)

Customer Experience has two goals: understand what causes a bad experience, and flag risky orders **at the moment the order is placed** so they can send a discount coupon. A bad experience is a rating of 1 or 2 stars.

Use binary logistic regression. Your analysis should show:

- whether the model as a whole and each predictor are significant.
- what each relevant factor means for the risk of a bad experience, interpreted through odds ratios.
- how well the model classifies orders it has not seen, using all the performance metrics we saw in class, each one interpreted.

Close with a clear answer for Customer Experience.

Part 5. Executive memo and decision log (5 points)

Executive memo. End your notebook with a memo to the COO, one page maximum. Answer the three questions with one clear recommendation for each leader, using numbers with units. The COO is not a statistician, so explain any technical term you use. Mention at least one limitation of your analysis.

Decision log (required). Start your notebook with a table of every important decision you made, the alternatives you considered and why you chose yours. A notebook without a decision log loses 10 points.

Grading rubric

The exam is worth 100. Most points are for interpretation and justified decisions, not for code that runs. The rules in "**What is expected in every analysis"** are graded inside every part.

Part	Pts	What is evaluated
1. Data preparation and exploration	15	- Data problems found and handled with clear justification. - Exploration that looks at how variables relate to each other and guides the rest of the analysis
2. Operations (regression)	35	- A justified model, tests with hypotheses and interpretation. - Assumptions checked and their consequences explained. - Effects interpreted in corresponding variables (business interpretation). - Evaluation on unseen data. - Clear answer for Operations (business terms).
3. Regional Director (ANOVA)	20	- Correct ANOVA. - Assumptions and pairwise comparisons between zones. - Well-supported comparison with the regression results. - Clear recommendation (business terms).
4. Customer Experience (logistic)	25	- Correct interpretation of odds ratios (interpreted in corresponding variables). - A model that can really be used when the order is placed. - Appropriate evaluation. - Clear answer for Customer Experience (business terms).
5. Executive memo	5	- Clear, non-technical answers to the three questions, with numbers and one limitation

Deductions. No decision log: −10 points. Notebook does not run from top to bottom: −5 points.